MPL equations and algorithm in matrix form for the Cox proportional hazard model

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This document is a complement to the article of Jun, Héritier and Lô (2013) describing in detail and in matrix form the "Newton multiplicative iterative" algorithm when approximating the baseline hazard function with a step function or with splines using a Gaussian or an Epanechnikov basis, or M-splines.

Notation

$\mathbf{A} = {}_{\mathbb{M}}\Big\{_{i=1}^{n} \; a_{ij} \; _{j=1}^{m}\Big\}$	an (n x m) matrix,
$\mathbf{a} = {}_{\mathbb{V}}\Big\{_{i=1}^{n} \; a_{i}\Big\}$	a vector of length n,
a	a scalar,
$\mathbf{A}^T, \mathbf{a}^T$	transposed matrix/vector,
$\mathbf{A}^{-1}$	inverse of $\mathbf{A}$,
$\langle\mathbf{a}\rangle$	squared diagonal matrix with vector $\mathbf{a}$ on the diagonal and 0 elsewhere,
$\mathrm{diag}(\mathbf{A})$	column vector of the diagonal elements of $\mathbf{A}$,
$\mathbf{A} \circ \mathbf{B}$	Hadamard (elementwise) product,
$\mathbf{A}/\mathbf{B}$	Hadamard (elementwise) division,
$\mathbf{A} \otimes \mathbf{B}$	Kronecker product,
$\log(\mathbf{A}) = {}_{\mathbb{M}}\Big\{_{i=1}^{n} \; \log(a_{ij}) \; _{j=1}^{m}\Big\}$	elementwise logarithm of $\mathbf{A}$,
$\mathbf{A}^{+} = {}_{\mathbb{M}}\Big\{_{i=1}^{n} \; a_{ij}\iota(a_{ij} > 0) \; _{j=1}^{m}\Big\}$	matrix $\mathbf{A}$ with its negative elements replaced by 0,
$\mathbf{A}^{\dot{+}} = {}_{\mathbb{M}}\Big\{_{i=1}^{n} \; a_{ij}^{\iota(a_{ij} \geqslant \epsilon)}\epsilon^{\iota(a_{ij} < \epsilon)} \; _{j=1}^{m}\Big\}$	matrix $\mathbf{A}$ with its elements smaller than ϵ replaced by ϵ, where $\epsilon > 0$,
$\mathbf{1}_n$	column vector of 1 of length n,
$\mathbf{I}_n = \langle\mathbf{1}_n\rangle$	(n x n) identity matrix,
$\iota(\cdot)$	indicator function which equal 1 if the condition holds and 0 otherwise.

1 Baseline and cumulative baseline equations

For the ith outcome, the Cox proportional hazard model is given by

$$h(t_i) = h_0(t_i)e^{\mathbf{x}_i^T\boldsymbol{\beta}}. \tag{1}$$

where:

- ▷ t_i denotes the ith outcome and $i = 1, ..., n$,
- ▷ $\mathbf{x}_i = (x_{i1}, ..., x_{ip})$ is the centred covariate vector corresponding to the ith observation,
- ▷ $\boldsymbol{\beta} = (\beta_1, ..., \beta_p)^T$ is the regression coefficient vector.
- ▷ $h_0(\cdot)$ is the baseline hazard function.

The baseline hazard $h_0(t)$ may be modelled using the following general expression [refer to Ma, Héritier and Lô (2013)]:

$$h_0(t) = \sum_{u=1}^{m} \theta_u \psi_u(t) \tag{2}$$

We will consider here two kinds of $\psi-$functions, the step function and the splines with a Gaussian basis, leading to different baseline hazard estimators.

1.1 Step function approximation

If the baseline hazard is approximated using a step function, the range of the data, $[t_{(1)}, t_{(n)}]$ is split in m sets, $\mathcal{S}_1, ..., \mathcal{S}_m$, such that $\bigcup_{u=1}^{m} \mathcal{S}_u = [t_{(1)}, t_{(n)}]$, $\mathcal{S}_u \bigcap \mathcal{S}_v = \varnothing$ if $u \neq v$.
$\psi_u(t)$ may be a uniform distribution with support $S_u = [\alpha_u, \alpha_{u+1}]$, leading to the following expressions of $\psi_u(t)$, $\Psi_u(t)$, and of the baseline and cumulative baseline hazard functions:

$$\psi_u(t) = \frac{\iota(\alpha_u \leqslant t < \alpha_{u+1})}{\alpha_{u+1} - \alpha_u}, \tag{3}$$

$$\Psi_u(t) = \int_{t_{(1)}}^{t} \psi_u(v)dv = \iota(\alpha_u \leqslant t) \left(\frac{t - \alpha_u}{\alpha_{u+1} - \alpha_u} \right)^{\iota(t<\alpha_{u+1})}, \tag{4}$$

$$h_0(t) = \sum_{u=1}^{m} \theta_u \psi_u(t) = \sum_{u=1}^{m} \theta_u \frac{\iota(\alpha_u \leqslant t < \alpha_{u+1})}{\alpha_{u+1} - \alpha_u}, \tag{5}$$

$$H_0(t) = \int_{t_{(1)}}^{t} h_0(v)dv = \sum_{u=1}^{m} \theta_u \Psi_u(t) = \sum_{u=1}^{m} \theta_u \iota(\alpha_u \leqslant t) \left(\frac{t - \alpha_u}{\alpha_{u+1} - \alpha_u} \right)^{\iota(t<\alpha_{u+1})}, \tag{6}$$

where $t_{(1)} \leqslant t \leqslant t_{(n)}$, $\alpha_u \in \mathbb{R}$, $\alpha_u \leqslant \alpha_{u+1}$ and $\Psi_u(\alpha_{u+1}) = 1$ for all $u = 1, ..., m$.
As the penalty function [refer to Section 2] influences the estimation of $\boldsymbol{\theta} = {}_{\mathbb{V}}\left\{_{u=1}^{m} \theta_u\right\}$, it is more appropriate, when approximating $h_0(t)$ with a step function, to define $\psi_u(t)$ such that $\theta_u = h_0(t)$ for $\alpha_u \leqslant t < \alpha_{u+1}$. Thus,

$$\psi_u(t) = \iota(\alpha_u \leqslant t < \alpha_{u+1}), \tag{7}$$

$$\Psi_u(t) = \int_{t_{(1)}}^{t} \psi_u(v)dv = \iota(\alpha_u \leqslant t) \left[t^{\iota(t<\alpha_{u+1})} \alpha_{u+1}^{\iota(\alpha_{u+1} \leqslant t)} - \alpha_u \right] \tag{8}$$

$$h_0(t) = \sum_{u=1}^{m} \theta_u \psi_u(t) = \sum_{u=1}^{m} \theta_u \iota(\alpha_u \leqslant t < \alpha_{u+1}), \tag{9}$$

$$H_0(t) = \int_{t_{(1)}}^{t} h_0(v)dv = \sum_{u=1}^{m} \theta_u \Psi_u(t) = \sum_{u=1}^{m} \theta_u \iota(\alpha_u \leqslant t) \left[t^{\iota(t<\alpha_{u+1})} \alpha_{u+1}^{\iota(\alpha_{u+1} \leqslant t)} - \alpha_u \right], \tag{10}$$

where $t_{(1)} \leqslant t \leqslant t_{(n)}$, $\alpha_u \in \mathbb{R}$ and $\alpha_u \leqslant \alpha_{u+1}$.
In their article, Jun, Héritier and Lô (2013) use the $\psi-$ function described in (8) but define $\Psi_u(t)$ and $H_0(t)$ in the following way:

$$\Psi_u(t) = \iota(\alpha_u \leqslant t)(\alpha_{u+1} - \alpha_u), \tag{11}$$

$$H_0(t) = \sum_{u=1}^{m} \theta_u \iota(\alpha_u \leqslant t)(\alpha_{u+1} - \alpha_u), \tag{12}$$

such that $\Psi_u(t) \neq \int_{t_{(1)}}^{t} \psi_u(v)dv$ and $H_0(t) \neq \int_{t_{(1)}}^{t} h_0(v)dv$.

1.2 Spline approximation using Gaussian bases

If the baseline hazard is approximated using splines with a Gaussian basis, $\psi_u(t)$ is a truncated Gaussian distribution with location parameter α_u, scale parameter σ_u and range $[t_{(1)}, t_{(n)}]$. This leads to the following expressions of $\psi_u(t)$, $\Psi_u(t)$, and of the baseline and cumulative baseline hazard functions:

$$\psi_u(t) = \frac{1}{\sigma_u \delta_u} \phi\left(\frac{t-\alpha_u}{\sigma_u}\right), \tag{13}$$

$$\Psi_u(t) = \int_{t_{(1)}}^{t} \psi_u(v)dv = \frac{1}{\delta_u}\left[\Phi\left(\frac{t-\alpha_u}{\sigma_u}\right) - \Phi\left(\frac{t_{(1)}-\alpha_u}{\sigma_u}\right)\right], \tag{14}$$

$$h_0(t) = \sum_{u=1}^{m} \theta_u \psi_u(t) = \sum_{u=1}^{m} \frac{\theta_u}{\sigma_u \delta_u} \phi\left(\frac{t-\alpha_u}{\sigma_u}\right), \tag{15}$$

$$H_0(t) = \int_{t_{(1)}}^{t} h_0(v)dv = \sum_{u=1}^{m} \theta_u \Psi_u(t) = \sum_{u=1}^{m} \frac{\theta_u}{\delta_u}\left[\Phi\left(\frac{t-\alpha_u}{\sigma_u}\right) - \Phi\left(\frac{t_{(1)}-\alpha_u}{\sigma_u}\right)\right], \tag{16}$$

where:

- ▷ $\phi(\cdot)$ and $\Phi(\cdot)$ respectively are the density and cumulative density functions of the standard Gaussian distribution,
- ▷ $t_{(1)} \leqslant t \leqslant t_{(n)}$,
- ▷ $\delta_u = \Phi\left(\frac{t_{(n)}-\alpha_u}{\sigma_u}\right) - \Phi\left(\frac{t_{(1)}-\alpha_u}{\sigma_u}\right)$,
- ▷ $\alpha_u \in \mathbb{R}$ and $\sigma_u > 0$.

1.3 M-spline approximation

If the baseline hazard is approximated using M-splines of order o, we get the following expressions of $\psi_u^o(t)$, $\Psi_u^o(t)$, and of the baseline and cumulative baseline hazard functions:

$$\psi_u^o(t) = \begin{cases} \frac{\iota(\alpha_u^\star \leqslant t < \alpha_{u+1}^\star)}{\alpha_{u+1}^\star - \alpha_u^\star} & \text{if } o = 1, \\ \frac{o}{o-1} \frac{\iota(\alpha_u^\star \leqslant t < \alpha_{u+o}^\star)}{\alpha_{u+o}^\star - \alpha_u^\star} \left[(t-\alpha_u^\star)\psi_u^{o-1}(t) + (\alpha_{u+o}^\star - t)\psi_{u+1}^{o-1}(t)\right] & \text{otherwise,} \end{cases} \tag{17}$$

$$\Psi_u^o(t) = \int_{-\infty}^{t} \psi_u^o(v)dv = \iota(\alpha_u^\star > t)\left[\sum_{v=u+1}^{\min(u+o,m+1)} \frac{\alpha_{v+o+1}^{\star\star} - \alpha_v^{\star\star}}{o+1} \psi_v^{o+1}(t)\right]^{\iota(\alpha_u^\star < t < \alpha_{u+o}^\star)}, \tag{18}$$

$$h_0(t) = \sum_{u=1}^{m} \theta_u \psi_u^o(t), \tag{19}$$

$$H_0(t) = \int_{t_{(1)}}^{t} h_0(v)dv = \sum_{u=1}^{m} \theta_u \Psi_u^o(t), \tag{20}$$

where:

- ▷ $t_{(1)} \leqslant t \leqslant t_{(n)}$,
- ▷ α_u the uth element of $\boldsymbol{\alpha}$, the knots vector of length n_α, $\alpha_u \in \mathbb{R}$ and $\alpha_u < \alpha_{u+1}$,
- ▷ $m = n_\alpha + o - 2$,
- ▷ $\boldsymbol{\alpha}^\star = \left[\min(\boldsymbol{\alpha})\mathbf{1}_{o-1}^T, \boldsymbol{\alpha}^T, \max(\boldsymbol{\alpha})\mathbf{1}_{o-1}^T\right]^T$,
- ▷ $\boldsymbol{\alpha}^{\star\star} = \left[\min(\boldsymbol{\alpha})\mathbf{1}_{o}^T, \boldsymbol{\alpha}^T, \max(\boldsymbol{\alpha})\mathbf{1}_{o}^T\right]^T$.

We will later only consider $o = 3$, i.e., M-splines of order 3, as the integration of $\psi_u^3(t)$ second derivative products, used to define (43), is easily obtained. We get the following nice M-spline integration properties: $\int_{-\infty}^{\alpha_u^\star} \psi_u^o(v)dv = 0$, $\int_{\alpha_u^\star}^{\alpha_{u+o}^\star} \psi_u^o(v)dv = 1$ and $\int_{\alpha_{u+o}^\star}^{\alpha_\infty^\star} \psi_u^o(v)dv = 0$.

1.4 Spline approximation using Epanechnikov bases

If the baseline hazard is approximated using splines with a Epanechnikov basis of "order" o, $\psi_u^o(t)$ is a (possibly truncated) Epanechnikov density function, and we get the following expressions of $\psi_u^o(t)$, $\Psi_u^o(t)$, and of the baseline and cumulative baseline hazard functions:

$$\psi_u^o(t) = \begin{cases} \frac{12\ \iota(\alpha_u^\star \leqslant t < \alpha_{u+o}^\star)(t-\alpha_{u+o}^\star)(t-2\alpha_u^\star+\alpha_{u+o}^\star)}{(2\alpha_u^\star-2\alpha_{u+o}^\star)^3} & \text{if } u=1, \\ \frac{6\ \iota(\alpha_u^\star \leqslant t < \alpha_{u+o}^\star)(t-\alpha_u^\star)(t-\alpha_{u+o}^\star)}{(\alpha_u^\star-\alpha_{u+o}^\star)^3} & \text{if } 1<u<m, \\ \frac{12\ \iota(\alpha_u^\star \leqslant t < \alpha_{u+o}^\star)(t-\alpha_u^\star)(t+\alpha_u^\star-2\alpha_{u+o}^\star)}{(2\alpha_u^\star-2\alpha_{u+o}^\star)^3} & \text{if } u=m, \end{cases} \tag{21}$$

$$\Psi_u^o(t) = \int\limits_{-\infty}^{t} \psi_u^o(v)dv = \begin{cases} \iota(\alpha_u^\star > t) \left[\frac{(t-\alpha_u^\star)(t^2-2t\alpha_u^\star-2\alpha_u^{\star 2}+6\alpha_u^\star\alpha_{u+o}^\star-3\alpha_{u+o}^{\star 2})}{2(\alpha_u^\star-\alpha_{u+o}^\star)^3}\right]^{\iota(\alpha_u^\star<t<\alpha_{u+o}^\star)} & u=1, \\ \iota(\alpha_u^\star > t) \left[\frac{(t-\alpha_u^\star)^2(2t+\alpha_u^\star-3\alpha_{u+o}^\star)}{(\alpha_u^\star-\alpha_{u+o}^\star)^3}\right]^{\iota(\alpha_u^\star<t<\alpha_{u+o}^\star)} & 1<u<m, \\ \iota(\alpha_u^\star > t) \left[\frac{(t-\alpha_u^\star)^2(t+2\alpha_u^\star-3\alpha_{u+o}^\star)}{2(\alpha_u^\star-\alpha_{u+o}^\star)^3}\right]^{\iota(\alpha_u^\star<t<\alpha_{u+o}^\star)} & u>1, \end{cases} \tag{22}$$

$$h_0(t) = \sum_{u=1}^{m} \theta_u \psi_u^o(t), \tag{23}$$

$$H_0(t) = \int\limits_{t_{(1)}}^{t} h_0(v)dv = \sum_{u=1}^{m} \theta_u \Psi_u^o(t), \tag{24}$$

where:

- ▷ $t_{(1)} \leqslant t \leqslant t_{(n)}$,
- ▷ α_u the uth element of $\boldsymbol{\alpha}$, the knots vector of length n_α, $\alpha_u \in \mathbb{R}$ and $\alpha_u < \alpha_{u+1}$,
- ▷ $m = n_\alpha + o - 2$,
- ▷ $\boldsymbol{\alpha}^\star = \left[\min(\boldsymbol{\alpha})\mathbf{1}_{o-1}^T, \boldsymbol{\alpha}^T, \max(\boldsymbol{\alpha})\mathbf{1}_{o-1}^T\right]^T$.

We get the following nice integration properties as with M-splines:
$\int\limits_{-\infty}^{\alpha_u^\star} \psi_u^o(v)dv = 0$, $\int\limits_{\alpha_u^\star}^{\alpha_{u+o}^\star} \psi_u^o(v)dv = 1$ and $\int\limits_{\alpha_{u+o}^\star}^{\alpha_\infty^\star} \psi_u^o(v)dv = 0$.

1.5 Spline approximation using cosine bases

If the baseline hazard is approximated using splines with a cosine basis of "order" o, $\psi_u^o(t)$ is a (possibly truncated) cosine density function, and we get the following expressions of $\psi_u^o(t)$, $\Psi_u^o(t)$, and of the baseline and cumulative baseline hazard functions:

$$\psi_u^o(t) = \begin{cases} \frac{\pi \ \iota(\alpha_u^\star \leqslant t < \alpha_{u+o}^\star) \ \text{Sin}\left[\frac{\pi(t-\alpha_{u+o}^\star)}{2\alpha_u^\star - 2\alpha_{u+o}^\star}\right]}{2(\alpha_{u+o}^\star - \alpha_u^\star)} & \text{if } u = 1, \\ \frac{\pi \ \iota(\alpha_u^\star \leqslant t < \alpha_{u+o}^\star) \ \text{Sin}\left[\frac{\pi(t-\alpha_{u+o}^\star)}{\alpha_u^\star - \alpha_{u+o}^\star}\right]}{2(\alpha_{u+o}^\star - \alpha_u^\star)} & \text{if } 1 < u < m, \\ \frac{\pi \ \iota(\alpha_u^\star \leqslant t < \alpha_{u+o}^\star) \ \text{Sin}\left[\frac{\pi(t+\alpha_u^\star - 2\alpha_{u+o}^\star)}{2\alpha_u^\star - 2\alpha_{u+o}^\star}\right]}{2(\alpha_{u+o}^\star - \alpha_u^\star)} & \text{if } u = m, \end{cases} \tag{25}$$

$$\Psi_u^o(t) = \int_{-\infty}^{t} \psi_u^o(v)dv = \begin{cases} \iota(\alpha_u^\star > t) \ \text{Sin}\left[\frac{\pi(t-3\alpha_u^\star+2\alpha_{u+o}^\star)}{2(\alpha_u^\star - \alpha_{u+o}^\star)}\right]^{\iota(\alpha_u^\star < t < \alpha_{u+o}^\star)} & u = 1, \\ \iota(\alpha_u^\star > t) \left\{\text{Sin}\left[\frac{\pi(t-3\alpha_u^\star+2\alpha_{u+o}^\star)}{2(\alpha_u^\star - \alpha_{u+o}^\star)}\right]^2\right\}^{\iota(\alpha_u^\star < t < \alpha_{u+o}^\star)} & 1 < u < m, \\ \iota(\alpha_u^\star > t) \left\{1 - \text{Sin}\left[\frac{\pi(t-4\alpha_u^\star+3\alpha_{u+o}^\star)}{2(\alpha_u^\star - \alpha_{u+o}^\star)}\right]\right\}^{\iota(\alpha_u^\star < t < \alpha_{u+o}^\star)} & u > 1, \end{cases} \tag{26}$$

$$h_0(t) = \sum_{u=1}^{m} \theta_u \psi_u^o(t), \tag{27}$$

$$H_0(t) = \int_{t_{(1)}}^{t} h_0(v)dv = \sum_{u=1}^{m} \theta_u \Psi_u^o(t), \tag{28}$$

where:

- ▷ $t_{(1)} \leqslant t \leqslant t_{(n)}$,
- ▷ α_u the uth element of $\boldsymbol{\alpha}$, the knots vector of length n_α, $\alpha_u \in \mathbb{R}$ and $\alpha_u < \alpha_{u+1}$,
- ▷ $m = n_\alpha + o - 2$,
- ▷ $\boldsymbol{\alpha}^\star = \left[\min(\boldsymbol{\alpha})\mathbf{1}_{o-1}^T, \boldsymbol{\alpha}^T, \max(\boldsymbol{\alpha})\mathbf{1}_{o-1}^T\right]^T$.

We get the following nice integration properties as with M-splines and Epanechnikov bases: $\int_{-\infty}^{\alpha_u^\star} \psi_u^o(v)dv = 0$, $\int_{\alpha_u^\star}^{\alpha_{u+o}^\star} \psi_u^o(v)dv = 1$ and $\int_{\alpha_{u+o}^\star}^{\alpha_\infty^\star} \psi_u^o(v)dv = 0$.

2 Penalised likelihood

The penalised log-likelihood of $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ given the fixed known or chosen elements, $\mathcal{L}_p(\boldsymbol{\beta}, \boldsymbol{\theta}|\mathbf{t}, \boldsymbol{\xi}, \mathbf{X}, \lambda, \boldsymbol{\Upsilon})$, is given by

$$\mathcal{L}_p(\boldsymbol{\beta}, \boldsymbol{\theta}|\mathbf{t}, \boldsymbol{\xi}, \mathbf{X}, \lambda, \boldsymbol{\Upsilon}) = \mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\theta}|\mathbf{t}, \boldsymbol{\xi}, \mathbf{X}) - \lambda \mathbf{J}(\boldsymbol{\theta}) \tag{29}$$

$$= \mathbf{1}_n^T \boldsymbol{\mathcal{L}} - \lambda \mathbf{J}(\boldsymbol{\theta}), \tag{30}$$

where:

- ▷ $\boldsymbol{\beta} = (\beta_1, ..., \beta_p)^T$ is the regression coefficient vector,
- ▷ $\boldsymbol{\theta} = (\theta 1, ..., \theta_m)^T$ is the Gaussian kernel coefficient vector,
- ▷ $\mathbf{t}$ is the outcome vector of length n,
- ▷ $\boldsymbol{\xi}$ is a vector of length n defined as following:

$$\boldsymbol{\xi} = {}_{\mathbb{V}}\left\{_{i=1}^{n} \ \xi_i\right\} = {}_{\mathbb{V}}\left\{_{i=1}^{n} \ \iota(i \in \mathcal{O})\right\}, \tag{31}$$

  where $\mathcal{O}$ denotes the set of fully observed (i.e., non censored) outcomes.
- ▷ $\mathbf{X}$ is the (n x p) centered covariate matrix,
- ▷ $\boldsymbol{\Upsilon}$ is the set of self-defined parameter vectors of the $\psi-$ and $\Psi-$ functions. When $\psi_u(t)$ and $\psi_v(t)$ are defined as in Section 1.1, $\boldsymbol{\Upsilon} = \boldsymbol{\alpha}$, where
  - ▷ $\boldsymbol{\alpha} = (\alpha_1, ..., \alpha_{m+1})^T$ is the vector of breaks of $[t_{(1)}, t_{(n)}]$ in m sets as described in Section 1.1,

  and when $\psi_u(t)$ and $\psi_v(t)$ are defined as in Section 1.2, $\boldsymbol{\Upsilon} = [\boldsymbol{\alpha}^T, \boldsymbol{\sigma}^T, \boldsymbol{\delta}^T]^T$, where
  - ▷ $\boldsymbol{\alpha} = (\alpha_1, ..., \alpha_m)^T$ is the Gaussian kernel location parameter vector of length m,

- ▷ $\boldsymbol{\sigma} = (\sigma_1, ..., \sigma_m)^T$ is the Gaussian kernel scale parameter vector of length m,
- ▷ $\boldsymbol{\delta} = (\delta_1, ..., \delta_m)^T$ is a vector of m constants forcing each truncated Gaussian basis to integrate to 1 on $[t_{(1)}, t_{(n)}]$.

▷ λ is the MPL smoothing parameter with $\lambda \in [0; \infty[$,

▷ $\boldsymbol{\mathcal{L}}$ is a vector of length n defined in the following way:

$$\boldsymbol{\mathcal{L}} = {}_{\mathbb{V}}\{_{i=1}^{n} \mathcal{L}_i\} = {}_{\mathbb{V}}\left\{_{i=1}^{n} \xi_i \log\left[\boldsymbol{\psi}(t_i)^T \boldsymbol{\theta} e^{\mathbf{x}_i^T \boldsymbol{\beta}}\right] - \boldsymbol{\Psi}(t_i)^T \boldsymbol{\theta} e^{\mathbf{x}_i^T \boldsymbol{\beta}}\right\} \tag{32}$$

$$= \boldsymbol{\xi} \circ \log\left[\boldsymbol{\psi}(t)\boldsymbol{\theta} \circ e^{\mathbf{X}\boldsymbol{\beta}}\right] - \boldsymbol{\Psi}(t)\boldsymbol{\theta} \circ e^{\mathbf{X}\boldsymbol{\beta}}, \tag{33}$$

where

$$\boldsymbol{\psi}(\mathbf{t}) = {}_{\mathbb{M}}\{_{i=1}^{n} \psi_u(t_i)_{u=1}^{m}\}, \tag{34}$$

$$\boldsymbol{\Psi}(\mathbf{t}) = {}_{\mathbb{M}}\{_{i=1}^{n} \Psi_u(t_i)_{u=1}^{m}\}, \tag{35}$$

$$\boldsymbol{\psi}_i(t_i) = {}_{\mathbb{V}}\{_{u=1}^{m} \psi_u(t_i)\}, \tag{36}$$

$$\boldsymbol{\Psi}_i(t_i) = {}_{\mathbb{V}}\{_{u=1}^{m} \Psi_u(t_i)\}, \tag{37}$$

$$\tag{38}$$

such that $h_0(\mathbf{t}) = \boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}$, $h_0(t_i) = \boldsymbol{\psi}(t_i)^T\boldsymbol{\theta}$, $H_0(\mathbf{t}) = \boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta}$, $H_0(t_i) = \boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta}$ and $\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\theta}|\mathbf{t}, \boldsymbol{\xi}, \mathbf{X}) = \mathbf{1}_n^T \boldsymbol{\mathcal{L}}$ is the (non-penalised) log-likelihood of $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$.

▷ $\mathbf{J}(\boldsymbol{\theta})$ is the penalty function.

We will later only consider quadratic penalty functions such that $\mathbf{J}(\boldsymbol{\theta}) = \boldsymbol{\theta}^T\mathbf{R}\boldsymbol{\theta}$ where $\mathbf{R}$ is a penalty matrix defined below. We will also re-express (30) using $\gamma = \frac{\lambda}{1+\lambda}$, a smoothing parameter belonging to the range $[0, 1[$. As $\lambda = \frac{\gamma}{1-\gamma}$, the $(1-\gamma$ scaled) penalised log-likelihood of $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ given the fixed known or chosen elements, $\mathcal{L}_p^\star(\boldsymbol{\beta}, \boldsymbol{\theta}|\mathbf{t}, \boldsymbol{\xi}, \mathbf{X}, \gamma, \boldsymbol{\Upsilon}, \mathbf{R})$, is given by

$$\mathcal{L}_p^\star(\boldsymbol{\beta}, \boldsymbol{\theta}|\mathbf{t}, \boldsymbol{\xi}, \mathbf{X}, \gamma, \boldsymbol{\Upsilon}, \mathbf{R}) = (1-\gamma)\mathcal{L}_p(\boldsymbol{\beta}, \boldsymbol{\theta}|\mathbf{t}, \boldsymbol{\xi}, \mathbf{X}, \gamma, \boldsymbol{\Upsilon}, \mathbf{R}) = (1-\gamma)\left(\mathbf{1}_n^T\boldsymbol{\mathcal{L}} - \frac{\gamma}{1-\gamma}\boldsymbol{\theta}^T\mathbf{R}\boldsymbol{\theta}\right), \tag{39}$$

$$= (1-\gamma)\mathbf{1}_n^T\boldsymbol{\mathcal{L}} - \gamma\boldsymbol{\theta}^T\mathbf{R}\boldsymbol{\theta}, \tag{40}$$

$$= \mathbf{1}_n^T\boldsymbol{\mathcal{L}}^\star - \gamma\boldsymbol{\theta}^T\mathbf{R}\boldsymbol{\theta}, \tag{41}$$

where $\mathbf{R}$ is the $(m \text{ x } m)$ roughness penalty matrix defined in next Section.

2.1 R penalty matrices

We describe here, when available, the first and second order penalties $\mathbf{R}$ penalty matrices corresponding to the baseline hazard approximations considered in Section 1.1 to 1.4. The first and second order penalty matrices, $\mathbf{R}_1$ and $\mathbf{R}_2$, are defined in the following way:

$$\mathbf{R}_1 = {}_{\mathbb{M}}\{_{u=1}^{m} r_{1_{uv}}{}_{v=1}^{m}\} = {}_{\mathbb{M}}\left\{_{u=1}^{m} \int_{t(1)}^{t(n)} \frac{\partial}{\partial t}\psi_u(t)\frac{\partial}{\partial t}\psi_v(t)dt_{v=1}^{m}\right\}, \tag{42}$$

$$\mathbf{R}_2 = {}_{\mathbb{M}}\{_{u=1}^{m} r_{2_{uv}}{}_{v=1}^{m}\} = {}_{\mathbb{M}}\left\{_{u=1}^{m} \int_{t(1)}^{t(n)} \frac{\partial^2}{\partial t\partial t}\psi_u(t)\frac{\partial^2}{\partial t\partial t}\psi_v(t)dt_{v=1}^{m}\right\}. \tag{43}$$

2.1.1 Step function approximation

When $\psi_u(t)$ and $\Psi_v(t)$ are defined as in Section 1.1, the following first and second order penalty matrices, $\mathbf{R}_1$ and $\mathbf{R}_2$, are used:

$$\mathbf{R}_1 = \begin{bmatrix} 1 & -1 & & & & & \\ -1 & 2 & -1 & & & & \\ & -1 & 2 & -1 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & -1 & 2 & -1 & \\ & & & & -1 & 2 & -1 \\ & & & & & -1 & 1 \end{bmatrix}, \tag{44}$$

$$\mathbf{R}_2 = \begin{bmatrix} 5 & -6 & 1 & & & & \\ -6 & 9 & -4 & 1 & & & \\ 1 & -4 & 6 & -4 & 1 & & \\ & \ddots & \ddots & \ddots & \ddots & \ddots & \\ & & 1 & -4 & 6 & -4 & 1 \\ & & & 1 & -4 & 9 & -6 \\ & & & & 1 & -6 & 5 \end{bmatrix}. \tag{45}$$

2.1.2 Spline approximation using Gaussian bases

When $\psi_u(t)$ and $\Psi_v(t)$ are defined as in Section 1.2, the elements $r_{1_{uv}}$ and $r_{2_{uv}}$ of $\mathbf{R}_1$ and $\mathbf{R}_2$ are defined in the following way:

$$r_{1_{uv}} = \frac{A_1(t_{(n)}) + B_1(t_{(n)})}{C_1(t_{(n)})} - \frac{A_1(t_{(1)}) + B_1(t_{(1)})}{C_1(t_{(1)})},$$
$$r_{2_{uv}} = \frac{A_2(t_{(n)}) + B_2(t_{(n)})}{C_2(t_{(n)})} - \frac{A_2(t_{(1)}) + B_2(t_{(1)})}{C_2(t_{(1)})},$$

where

$$A_1(t) = \left(\sigma_u^2 + \sigma_v^2\right)^{\frac{1}{2}} \left[(\alpha_u - t)\sigma_u^2 + (\alpha_v - t)\sigma_v^2\right] \phi\left(\frac{t-\alpha_u}{\sigma_u}\right) \phi\left(\frac{t-\alpha_v}{\sigma_v}\right),$$
$$B_1(t) = \sigma_u\sigma_v \left[\sigma_u^2 + \sigma_2^2 - (\alpha_u - \alpha_v)^2\right] \ \phi\left[\frac{\alpha_v - \alpha_u}{(\sigma_u^2 + \sigma_v^2)^{\frac{1}{2}}}\right] \left\{\Phi\left[\frac{(t-\alpha_v)\sigma_u^2 + (t-\alpha_u)\sigma_v^2}{\sigma_u\sigma_v(\sigma_u^2+\sigma_v^2)^{\frac{1}{2}}}\right] - \frac{1}{2}\right\},$$
$$C_1(t) = \sigma_u\sigma_v(\sigma_u^2 + \sigma_v^2)^{\frac{5}{2}}\delta_u\delta_v,$$

and

$$\begin{aligned} A_2(t) = {} & \left(\sigma_u^2 + \sigma_v^2\right)^{\frac{1}{2}} \phi\left(\frac{t-\alpha_u}{\sigma_u}\right) \phi\left(\frac{t-\alpha_v}{\sigma_v}\right) \Big\{\sigma_u^6(\alpha_v - t)\left[(\alpha_u - t)^2 - \sigma_u^2\right] \\ & + \sigma_u^4\sigma_v^2\left[(t - 4\alpha_v + 3\alpha_u)\sigma_u^2 - (\alpha_u - t)^2(3t - 4\alpha_u + \alpha_v)\right] - \sigma_u^2\sigma_v^4(\alpha_v - t)^2(3t - 4\alpha_u + \alpha_v) \\ & + \sigma_v^6\left[(t + 3\alpha_v - 4\alpha_u)\sigma_u^2 - (\alpha_v - t)^2(t - \alpha_u)\right] + \sigma_v^8(t - \alpha_u)\Big\}, \\ B_2(t) = {} & \sigma_u^3\sigma_v^3 \ \phi\left[\frac{\alpha_v - \alpha_u}{(\sigma_u^2 + \sigma_v^2)^{\frac{1}{2}}}\right] \left\{\Phi\left[\frac{(t-\alpha_v)\sigma_u^2 + (t-\alpha_u)\sigma_v^2}{\sigma_u\sigma_v(\sigma_u^2+\sigma_v^2)^{\frac{1}{2}}}\right] - \frac{1}{2}\right\} \Big\{\alpha_v^4 + \alpha_u^4 - 4\alpha_v^3\alpha_u \\ & + 6\alpha_v^2(\alpha_u^2 - \sigma_u^2 - \sigma_v^2) - 6\alpha_u^2(\sigma_u^2 + \sigma_v^2) - 4\alpha_u\alpha_v\left[\alpha_u^2 - 3(\sigma_u^2 + \sigma_v^2)\right] + 3(\sigma_u^2 + \sigma_v^2)^2\Big\}, \\ C_2(t) = {} & \sigma_u^3\sigma_v^3(\sigma_u^2 + \sigma_v^2)^{\frac{9}{2}}\delta_u\delta_v. \end{aligned}$$

2.1.3 M-spline approximation

When $\psi_u^o(t)$ and $\Psi_v^o(t)$ are defined as in Section 1.3, the $\mathbf{R}$ penalty matrix is (easily) defined only for penalty of order $o-1$. Thus, $\mathbf{R}_1$ is available when using M-splines of 2nd order ($o=2$), and $\mathbf{R}_2$ is available when using M-splines of 3rd order ($o=3$). The element $r_{o-1_{uv}}$ of $\mathbf{R}_{o-1}$ is defined in the following way:

$$r_{o-1_{uv}} = \begin{cases} 0 & \text{if } \alpha_{u+o}^\star < \alpha_v^\star, \\ & \text{otherwise,} \end{cases}$$

where $\alpha_u^\star$ is the uth element of $\boldsymbol{\alpha}^\star$, as defined in Section 1.3.

2.1.4 Spline approximation using Epanechnikov bases

When $\psi_u^o(t)$ and $\Psi_v^o(t)$ are defined as in Section 1.4, the elements $r_{1_{uv}}$ and $r_{2_{uv}}$ of $\mathbf{R}_1$ and $\mathbf{R}_2$ are defined in the following way:

$$r_{1_{uv}} = \begin{cases} \frac{3(\alpha^\star_{u+o}-\alpha^\star_v)\left[\alpha^{\star 2}_v+\alpha^{\star 2}_{u+o}+\alpha^\star_v\alpha^\star_{u+o}+3\alpha^\star_u\left(\alpha^\star_u-\alpha^\star_{u+o}-\alpha^\star_v\right)\right]}{\left(\alpha^\star_{u+o}-\alpha^\star_u\right)^6} & \text{if } (u=1 \text{ or } v=1) \text{ and } \alpha^\star_v < \alpha^\star_{u+o}, \\ \frac{6\left[\left(\alpha^\star_{u+o}+\alpha^\star_u-2\alpha^\star_v\right)^3-\left(\alpha^\star_{u+o}-\alpha^\star_u\right)^3\right]}{\left(\alpha^\star_{u+o}-\alpha^\star_u\right)^6} & \text{if } 1<u<m \text{ and } \alpha^\star_v < \alpha^\star_{u+o}, \\ \frac{3(\alpha^\star_{u+o}-\alpha^\star_v)^3}{\left(\alpha^\star_{u+o}-\alpha^\star_u\right)^6} & \text{if } (u=m \text{ or } v=m) \text{ and } \alpha^\star_v < \alpha^\star_{u+o}, \\ 0 & \text{otherwise,} \end{cases}$$

$$r_{2_{uv}} = \begin{cases} \frac{4^{\iota(1<u<m)}4^{\iota(1<v<m)}9\left(\alpha^\star_{u+o}-\alpha^\star_v\right)}{\left(\alpha^\star_{u+o}-\alpha^\star_u\right)^3\left(\alpha^\star_{v+o}-\alpha^\star_v\right)^3} & \text{if } \alpha^\star_v < \alpha^\star_{u+o}, \\ 0 & \text{otherwise,} \end{cases}$$

where $\alpha^\star_u$ is the uth element of $\boldsymbol{\alpha}^\star$, as defined in Section 1.4.

2.1.5 Spline approximation using cosine bases

When $\psi_u^o(t)$ and $\Psi_v^o(t)$ are defined as in Section 1.5, the elements $r_{1_{uv}}$ and $r_{2_{uv}}$ of $\mathbf{R}_1$ and $\mathbf{R}_2$ are defined in the following way:

$$r_{1_{uv}} = \begin{cases} \frac{\pi^3\left\{2\pi\left(\alpha^\star_{u+o}-\alpha^\star_v\right)+\left(\alpha^\star_{u+o}-\alpha^\star_u\right)\,\mathrm{Sin}\left[\frac{2\pi\left(\alpha^\star_{u+o}-\alpha^\star_v\right)}{\alpha^\star_{u+o}-\alpha^\star_u}\right]\right\}}{4^{\iota(1<u<m)}4^{\iota(1<v<m)}4\left(\alpha^\star_{u+o}-\alpha^\star_u\right)^4} & \text{if } \alpha^\star_v < \alpha^\star_{u+o}, \\ 0 & \text{otherwise,} \end{cases}$$

$$r_{2_{uv}} = \begin{cases} 0 & \text{if } \alpha^\star_{u+o} < \alpha^\star_v, \\ \frac{\pi^5}{2^{\iota(1<u<m)}2^{\iota(1<v<m)}4\left(\alpha^\star_{u+o}-\alpha^\star_u\right)^2\left(\alpha^\star_{v+o}-\alpha^\star_v\right)^2\left[\left(\alpha^\star_{u+o}-\alpha^\star_u\right)^2-\left(\alpha^\star_{v+o}-\alpha^\star_v\right)^2\right]} & \\ \times\left\{2\left(\alpha^\star_{v+o}-\alpha^\star_v\right)\mathrm{Sin}\left[\frac{\pi\left(\alpha^\star_{v+o}-\alpha^\star_{u+o}\right)}{\alpha^\star_v-\alpha^\star_{v+o}}\right]\right. & \\ \quad+\left(\alpha^\star_{v+o}-\alpha^\star_v+\alpha^\star_u-\alpha^\star_{u+o}\right)\mathrm{Sin}\left[\frac{\pi\left(\alpha^\star_v-\alpha^\star_{u+o}\right)}{\alpha^\star_u-\alpha^\star_{u+o}}\right] & \\ \quad\left.+\left(\alpha^\star_v-\alpha^\star_{v+o}+\alpha^\star_u-\alpha^\star_{u+o}\right)\mathrm{Sin}\left[\frac{\pi\left(\alpha^\star_v-2\alpha^\star_u+\alpha^\star_{u+o}\right)}{\alpha^\star_u-\alpha^\star_{u+o}}\right]\right\} & \text{otherwise,} \end{cases}$$

where $\alpha^\star_u$ is the uth element of $\boldsymbol{\alpha}^\star$, as defined in Section 1.4.

3 Estimation

Optimisation of the penalised log-likelihood given in (30) is achieved by using the Newton multiplicative-algorithm described in Ma, Héritier and Lô (2013), which requires the first and second derivatives of the objective function by $\boldsymbol{\beta}$ and its first derivative by $\boldsymbol{\theta}$. Considering quadratic penalty functions of form $\mathbf{J}(\boldsymbol{\theta}) = \boldsymbol{\theta}^T \mathbf{R} \boldsymbol{\theta}$, these later are given by [refer to Section A.1 to A.3 of the appendix]:

$$\frac{\partial}{\partial \boldsymbol{\beta}} \mathcal{L}_p^{\star} = \boldsymbol{\nabla}_{\boldsymbol{\beta}} = (1-\gamma)\, \mathbf{X}^T \left[\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t}) \boldsymbol{\theta} \circ e^{\mathbf{X}\boldsymbol{\beta}} \right], \tag{46}$$

$$\frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}} \mathcal{L}_p^{\star} = \boldsymbol{\mathcal{H}}_{\boldsymbol{\beta}} = (\gamma-1)\, \mathbf{X}^T \left\langle \boldsymbol{\Psi}(t) \boldsymbol{\theta} \circ e^{\mathbf{X}\boldsymbol{\beta}} \right\rangle \mathbf{X} \tag{47}$$

$$\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L}_p^{\star} = \boldsymbol{\nabla}_{\boldsymbol{\theta}} = (1-\gamma) \left\{ \boldsymbol{\psi}(\mathbf{t})^T \left[\boldsymbol{\xi} / \boldsymbol{\psi}(t) \boldsymbol{\theta} \right] - \boldsymbol{\Psi}(\mathbf{t})^T e^{\mathbf{X}\boldsymbol{\beta}} \right\} - 2\gamma \mathbf{R} \boldsymbol{\theta}. \tag{48}$$

Until convergence, the $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ parameters are updated in the following way:

$$\boldsymbol{\beta}^{(k)} = \boldsymbol{\beta}^{(k-1)} + \omega^{(k)} \boldsymbol{\mathcal{H}}_{\boldsymbol{\beta}}^{(k)^{-1}} \boldsymbol{\nabla}_{\boldsymbol{\beta}}^{(k)} \text{ and then} \tag{49}$$

$$\boldsymbol{\theta}^{(k)} = \boldsymbol{\theta}^{(k-1)} + \nu^{(k)} \mathbf{s}^{(k)^T} \boldsymbol{\nabla}^{(\boldsymbol{k})}{}_{\boldsymbol{\theta}}, \tag{50}$$

$$\tag{51}$$

where k denotes the kth iteration and

$$\boldsymbol{\nabla}_{\boldsymbol{\beta}}^{(k)} = (1-\gamma)\, \mathbf{X}^T \left[\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t}) \boldsymbol{\theta}^{(k-1)} \circ e^{\mathbf{X}\boldsymbol{\beta}^{(k-1)}} \right], \tag{52}$$

$$\boldsymbol{\mathcal{H}}_{\boldsymbol{\beta}}^{(k)} = (\gamma-1)\, \mathbf{X}^T \left\langle \boldsymbol{\Psi}(t) \boldsymbol{\theta}^{(k-1)} \circ e^{\mathbf{X}\boldsymbol{\beta}^{(k-1)}} \right\rangle \mathbf{X}, \tag{53}$$

$$\boldsymbol{\nabla}_{\boldsymbol{\theta}}^{(k)} = (1-\gamma) \left\{ \boldsymbol{\psi}(\mathbf{t})^T \left[\boldsymbol{\xi} / \boldsymbol{\psi}(t) \boldsymbol{\theta}^{(k-1)} \right] - \boldsymbol{\Psi}(\mathbf{t})^T e^{\mathbf{X}\boldsymbol{\beta}^{(k)}} \right\} - 2\gamma \mathbf{R} \boldsymbol{\theta}^{(k-1)}, \tag{54}$$

$$\mathbf{s}^{(k)} = \boldsymbol{\theta}^{(k-1)} / \left\{ (1-\gamma) \boldsymbol{\Psi}(\mathbf{t})^T e^{\mathbf{X}\boldsymbol{\beta}^{(k)}} + \left[2\gamma \mathbf{R} \boldsymbol{\theta}^{(k-1)} \right]^+ + \epsilon \mathbf{1}_m \right\}. \tag{55}$$

4 Implementation

Input:

- ▷ $\mathbf{t}$, the outcome vector of length n,
- ▷ $\boldsymbol{\xi}$, a vector of length n defined in (31),
- ▷ $\mathbf{X}$ and $\mathbf{X}^{\dagger}$ the (n x p) centred and non-centered covariate matrices,
- ▷ $\mathbf{R}$, an (m x m) matrix defined in (44) to (43),
- ▷ γ, the MPL smoothing parameter with $0 < \gamma < 1$,
- ▷ $\kappa > 1$, a constant used to decrease the Newton step length when necessary,
- ▷ $\epsilon > 0$, a (very small) constant used to avoid divisions by 0 or by to small values (like 10^{-300}),
- ▷ the (n x m) $\boldsymbol{\psi}-$ and $\boldsymbol{\Psi}-$ matrices:
  - ▷ $\boldsymbol{\psi}(\mathbf{t}) = {}_{\mathbb{M}}\left\{_{i=1}^{n} \quad \psi_u(t,\boldsymbol{\Upsilon}) \quad _{u=1}^{m}\right\}$,
  - ▷ $\boldsymbol{\Psi}(\mathbf{t}) = {}_{\mathbb{M}}\left\{_{i=1}^{n} \quad \Psi_u(t,\boldsymbol{\Upsilon}) \quad _{u=1}^{m}\right\}$,

  where
  - ▷ $\psi-$ and $\Psi-$ functions are defined in Sections 1.1 and 1.2,
  - ▷ $\boldsymbol{\Upsilon}$ is the set of self-defined parameter vectors of the $\psi-$ and $\Psi-$ functions:
    When $\psi_u(t)$ and $\Psi_v(t)$ are defined as in Section 1.1, $\boldsymbol{\Upsilon} = \boldsymbol{\alpha}$, where
    - ▷ $\boldsymbol{\alpha} = (\alpha_1, ..., \alpha_{m+1})^T$ is the vector of breaks of $[t_{(1)}, t_{(n)}]$ in m sets as described in Section 1.1.

    When $\psi_u(t)$ and $\Psi_v(t)$ are defined as in Section 1.2, $\boldsymbol{\Upsilon} = [\boldsymbol{\alpha}^T, \boldsymbol{\sigma}^T, \boldsymbol{\delta}^T]^T$, where
    - ▷ $\boldsymbol{\alpha} = (\alpha_1, ..., \alpha_m)^T$ is the location parameter vector of of the m truncated Gaussian densities,
    - ▷ $\boldsymbol{\sigma} = (\sigma_1, ..., \sigma_m)^T$ is the scale parameter vector of of the m truncated Gaussian densities.
    - ▷ $\boldsymbol{\delta} = (\delta_1, ..., \delta_m)^T$ is a vector of m constants forcing each Gaussian basis to integrate to 1 on $[t_{(1)}, t_{(n)}]$.

Output:

- ▷ $\widehat{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{\theta}}$, the MPL model estimates,
- ▷ $\widehat{\mathcal{L}}_p$, the value of the penalised log-likelihood at convergence.

Initialise:

- ▷ $\boldsymbol{\beta}^{(0)} = {}_{\mathbb{V}}\left\{_{j=1}^{p} \ \beta_j\right\} = \mathbf{1}_p$,
- ▷ $\boldsymbol{\theta}^{(0)} = {}_{\mathbb{V}}\left\{_{u=1}^{m} \ \theta_u\right\} = \mathbf{1}_m$,
- ▷ $\boldsymbol{\mu}^{(0)} = {}_{\mathbb{V}}\left\{_{i=1}^{n} \ \mu_i\right\} = e^{\mathbf{X}\boldsymbol{\beta}^{(0)}}$,
- ▷ $l_p^{(0)} = (1-\gamma)\left\{\boldsymbol{\xi}^T \log\left(\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}^{(0)} \circ \boldsymbol{\mu}^{(0)}\right) - \mathbf{1}_n^T\left[\boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta}^{(0)} \circ \boldsymbol{\mu}^{(0)}\right]\right\} - \gamma\boldsymbol{\theta}^{(0)^T}\mathbf{R}\boldsymbol{\theta}^{(0)}$,
- ▷ $\varepsilon = 10^6$,
- ▷ $k = 0$.

Remark:
For algorithmic reasons, all matrix products of form $\mathbf{XAY}$, where $\mathbf{A}$ is a diagonal matrix, are re-expressed as $\mathbf{X}^{\star}\mathbf{Y}$ or $\mathbf{X}\mathbf{Y}^{\star}$, where $\mathbf{X}^{\star}$ and $\mathbf{Y}^{\star}$ are re-scaled using elements of diag($\mathbf{A}$), such that

$$\mathbf{XAY} = \mathbf{X}^{\star}\mathbf{Y} = \mathbf{X}\mathbf{Y}^{\star}.$$

For example,

$$\begin{aligned}\frac{\partial^2}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}}\mathcal{L}_p^{\star} = \mathcal{H}_{\boldsymbol{\beta}} &= (\gamma-1)\,\mathbf{X}^T\left\langle\boldsymbol{\Psi}(t)\boldsymbol{\theta} \circ e^{\mathbf{X}\boldsymbol{\beta}}\right\rangle\mathbf{X} \\ &= (\gamma-1)\left(\mathbf{X} \circ \left\{\mathbf{1}_p^T \otimes \left[\boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta} \circ e^{\mathbf{X}\boldsymbol{\beta}}\right]\right\}\right)^T\mathbf{X}.\end{aligned}$$

Estimation algorithm:

begin

while $\varepsilon > 10^{-20}$ **do**

$k \leftarrow k + 1;$

update $\boldsymbol{\beta}$:

$\omega^{(k)} \leftarrow 1;$

$\nabla_{\boldsymbol{\beta}}^{(k)} \leftarrow (1-\gamma)\,\mathbf{X}^T\left[\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta}^{(k-1)} \circ \boldsymbol{\mu}^{(k-1)}\right];$

$\mathcal{H}_{\boldsymbol{\beta}}^{(k)} \leftarrow (1-\gamma)\left(\mathbf{X} \circ \left\{\mathbf{1}_p^T \otimes \left[\boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta}^{(k-1)} \circ \boldsymbol{\mu}^{(k-1)}\right]\right\}\right)^T \mathbf{X};$

$\blacktriangledown_{\boldsymbol{\beta}}^{(k)} \leftarrow \left(\mathcal{H}_{\boldsymbol{\beta}}^{(k)}\right)^{-1} \nabla_{\boldsymbol{\beta}}^{(k)};$

$\boldsymbol{\beta}^{(k)} \leftarrow \boldsymbol{\beta}^{(k-1)} + \omega^{(k)} \blacktriangledown_{\boldsymbol{\beta}}^{(k)};$

$\boldsymbol{\mu}^{(k)} \leftarrow e^{\mathbf{X}\boldsymbol{\beta}^{(k)}};$

$\mathcal{L}_p^{\star(k-\frac{1}{2})} \leftarrow (1-\gamma)\left\{\boldsymbol{\xi}^T \log\left(\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}^{(k-1)} \circ \boldsymbol{\mu}^{(k)}\right) - \mathbf{1}_n^T\left[\boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta}^{(k-1)} \circ \boldsymbol{\mu}^{(k)}\right]\right\} - \gamma \boldsymbol{\theta}^{(k-1)^T} \mathbf{R} \boldsymbol{\theta}^{(k-1)};$

if $\mathcal{L}_p^{\star(k-\frac{1}{2})} < \mathcal{L}_p^{\star(k-1)}$ **then**

while $\mathcal{L}_p^{\star(k-\frac{1}{2})} < \mathcal{L}_p^{\star(k-1)}$ **do**

$\omega^{(k)} \leftarrow \frac{\omega^{(k)}}{\kappa};$

$\boldsymbol{\beta}^{(k)} \leftarrow \boldsymbol{\beta}^{(k-1)} + \omega^{(k)} \blacktriangledown_{\boldsymbol{\beta}}^{(k)};$

$\boldsymbol{\mu}^{(k)} \leftarrow e^{\mathbf{X}\boldsymbol{\beta}^{(k)}};$

$\mathcal{L}_p^{\star(k-\frac{1}{2})} \leftarrow (1-\gamma)\left\{\boldsymbol{\xi}^T \log\left(\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}^{(k-1)} \circ \boldsymbol{\mu}^{(k)}\right) - \mathbf{1}_n^T\left[\boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta}^{(k-1)} \circ \boldsymbol{\mu}^{(k)}\right]\right\}$
$\qquad - \gamma \boldsymbol{\theta}^{(k-1)^T} \mathbf{R} \boldsymbol{\theta}^{(k-1)};$

update $\boldsymbol{\theta}$:

$\nu^{(k)} \leftarrow 1;$

$\nabla_{\boldsymbol{\theta}}^{(k)} \leftarrow (1-\gamma)\left(\left\{\boldsymbol{\psi}(\mathbf{t}) / \left[\mathbf{1}_m^T \otimes \boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}^{(k-1)}\right]\right\}^T \boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t})^T \boldsymbol{\mu}^{(k)}\right) - 2\gamma \mathbf{R}\boldsymbol{\theta}^{(k-1)};$

$\mathbf{s}^{(k)} \leftarrow \boldsymbol{\theta}^{(k-1)} / \left\{(1-\gamma)\boldsymbol{\Psi}(\mathbf{t})^T \boldsymbol{\mu}^{(k)} + \left[2\gamma \mathbf{R}\boldsymbol{\theta}^{(k-1)}\right]^+ + \epsilon \mathbf{1}_m\right\};$

$\blacktriangledown_{\boldsymbol{\theta}}^{(k)} \leftarrow \mathbf{s}^{(k)^T} \circ \nabla_{\boldsymbol{\theta}}^{(k)};$

$\boldsymbol{\theta}^{(k)} \leftarrow \left(\boldsymbol{\theta}^{(k-1)} + \nu^{(k)} \blacktriangledown_{\boldsymbol{\beta}}^{(k)}\right)^{\ddagger};$

$\mathcal{L}_p^{\star(k)} \leftarrow (1-\gamma)\left\{\boldsymbol{\xi}^T \log\left(\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}^{(k)} \circ \boldsymbol{\mu}^{(k)}\right) - \mathbf{1}_n^T\left[\boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta}^{(k)} \circ \boldsymbol{\mu}^{(k)}\right]\right\} - \gamma \boldsymbol{\theta}^{(k)^T} \mathbf{R} \boldsymbol{\theta}^{(k)};$

if $\mathcal{L}_p^{\star(k)} < \mathcal{L}_p^{\star(k-\frac{1}{2})}$ **then**

while $\mathcal{L}_p^{\star(k)} < \mathcal{L}_p^{\star(k-\frac{1}{2})}$ **do**

$\nu^{(k)} \leftarrow \frac{\nu^{(k)}}{\kappa};$

$\boldsymbol{\theta}^{(k)} \leftarrow \left(\boldsymbol{\theta}^{(k-1)} + \nu^{(k)} \blacktriangledown_{\boldsymbol{\beta}}^{(k)}\right)^{\ddagger};$

$\mathcal{L}_p^{\star(k)} \leftarrow (1-\gamma)\left\{\boldsymbol{\xi}^T \log\left(\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}^{(k)} \circ \boldsymbol{\mu}^{(k)}\right) - \mathbf{1}_n^T\left[\boldsymbol{\Psi}(\mathbf{t})\boldsymbol{\theta}^{(k)} \circ \boldsymbol{\mu}^{(k)}\right]\right\} - \gamma \boldsymbol{\theta}^{(k)^T} \mathbf{R} \boldsymbol{\theta}^{(k)};$

update ε:

$\varepsilon \leftarrow \max\left(\mathbf{1}_n^T \left|\boldsymbol{\beta}^{(k)} - \boldsymbol{\beta}^{(k-1)}\right| + \mathbf{1}_m^T \left|\boldsymbol{\theta}^{(k)} - \boldsymbol{\theta}^{(k-1)}\right|\right);$

$\widehat{\boldsymbol{\beta}} \leftarrow \boldsymbol{\beta}^{(k)};$

$\widehat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}^{(k)} e^{\frac{1}{n}\mathbf{1}_n \mathbf{X}^{\dagger}};$

5 Inference

Two cases, depending on values of $\lambda = \frac{\gamma}{1-\gamma}$, are considered.

5.1 Asymptotics when $\frac{\gamma}{1-\gamma} \to 0$

When $\frac{\gamma}{1-\gamma} \to 0$, the penalty disappears and, under some regularity conditions, the asymptotic covariance matrix of $\boldsymbol{\eta} = (\boldsymbol{\beta}^T, \boldsymbol{\theta}^T)^T$, $\mathbf{V}(\boldsymbol{\eta})$, equals $(-\boldsymbol{\mathcal{H}})^{-1}(\boldsymbol{\eta})$, where $\boldsymbol{\mathcal{H}}$ is the hessian matrix of the unpenalised likelihood function. The estimated asymptotic covariance matrix of $\widehat{\boldsymbol{\eta}} = (\widehat{\boldsymbol{\beta}}^T, \widehat{\boldsymbol{\theta}}^T)^T$, $\widehat{\mathbf{V}}(\widehat{\boldsymbol{\eta}})$, is then obtained by pluggin-in $\widehat{\boldsymbol{\eta}}$ for $\boldsymbol{\eta}$ and using the empirical version of $\boldsymbol{\mathcal{H}}(\boldsymbol{\eta})$. We then have

$$\sqrt{n}(\widehat{\boldsymbol{\eta}} - \boldsymbol{\eta}) \sim \text{MVN}\left[\mathbf{0}_{p+m}, (-\widehat{\boldsymbol{\mathcal{H}}})^{-1}(\widehat{\boldsymbol{\eta}})\right], \tag{56}$$

where MVN() denotes the multivariate normal distribution and

$$-\widehat{\boldsymbol{\mathcal{H}}}(\widehat{\boldsymbol{\eta}}) = -\begin{bmatrix} \widehat{\boldsymbol{\mathcal{H}}}_1(\widehat{\boldsymbol{\eta}}) & \widehat{\boldsymbol{\mathcal{H}}}_3(\widehat{\boldsymbol{\eta}}) \\ \widehat{\boldsymbol{\mathcal{H}}}_2(\widehat{\boldsymbol{\eta}}) & \widehat{\boldsymbol{\mathcal{H}}}_4(\widehat{\boldsymbol{\eta}}) \end{bmatrix} \tag{57}$$

$$= -\left.\begin{bmatrix} \frac{\partial^2}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^T}\boldsymbol{\mathcal{L}}^\star & \frac{\partial^2}{\partial\boldsymbol{\beta}\partial\boldsymbol{\theta}^T}\boldsymbol{\mathcal{L}}^\star \\ \frac{\partial^2}{\partial\boldsymbol{\theta}\partial\boldsymbol{\beta}^T}\boldsymbol{\mathcal{L}}^\star & \frac{\partial^2}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}\boldsymbol{\mathcal{L}}^\star \end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}} \tag{58}$$

$$= \begin{bmatrix} (1-\gamma)\mathbf{X}^T\left\langle \boldsymbol{\Psi}(t)\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right\rangle \mathbf{X} & (1-\gamma)\mathbf{X}^T\langle e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\rangle\, \boldsymbol{\Psi}(\mathbf{t}) \\ (1-\gamma)\boldsymbol{\Psi}(\mathbf{t})^T\langle e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\rangle\, \mathbf{X} & (1-\gamma)\boldsymbol{\psi}(\mathbf{t})^T\left\langle \boldsymbol{\xi}/[\boldsymbol{\psi}(\mathbf{t})\widehat{\boldsymbol{\theta}}]^2\right\rangle \boldsymbol{\psi}(\mathbf{t}) \end{bmatrix}. \tag{59}$$

As $\frac{\gamma}{1-\gamma} \to 0$, we have $\gamma \to 0$ such that

$$-\widehat{\boldsymbol{\mathcal{H}}}(\widehat{\boldsymbol{\eta}}) = \begin{bmatrix} \mathbf{X}^T\left\langle \boldsymbol{\Psi}(t)\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right\rangle \mathbf{X} & \mathbf{X}^T\langle e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\rangle\, \boldsymbol{\Psi}(\mathbf{t}) \\ \boldsymbol{\Psi}(\mathbf{t})^T\langle e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\rangle\, \mathbf{X} & \boldsymbol{\psi}(\mathbf{t})^T\left\langle \boldsymbol{\xi}/[\boldsymbol{\psi}(\mathbf{t})\widehat{\boldsymbol{\theta}}]^2\right\rangle \boldsymbol{\psi}(\mathbf{t}) \end{bmatrix}. \tag{60}$$

Refer to the appendix for detail. Note that $\widehat{\boldsymbol{\mathcal{H}}}_3(\widehat{\boldsymbol{\eta}})$ in (57) is the sample estimate of $\boldsymbol{\mathcal{H}}_{\boldsymbol{\beta}}$ in (47).
When deducing the standard deviations of $\widehat{\boldsymbol{\eta}}$ from (56), the standard deviation of $\widehat{\boldsymbol{\theta}}$ depends on $\widehat{\boldsymbol{\mathcal{H}}}_3(\widehat{\boldsymbol{\eta}})$ and $\widehat{\boldsymbol{\mathcal{H}}}_4(\widehat{\boldsymbol{\eta}})$, the second being (supposedly) more important than the first. Thus, when the baseline hazard is approximated by a step function as in (8), we have

$$\widehat{\boldsymbol{\sigma}}^2_{\widehat{\boldsymbol{\theta}}} \cong \left\{\text{diag}\left[\boldsymbol{\psi}(\mathbf{t})^T\left\langle \boldsymbol{\xi}/[\boldsymbol{\psi}(\mathbf{t})\widehat{\boldsymbol{\theta}}]^2\right\rangle \boldsymbol{\psi}(\mathbf{t})\right]\right\}^{-1} \tag{61}$$

$$\cong \frac{\widehat{\boldsymbol{\theta}}^2}{\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\xi}}, \tag{62}$$

where $\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\xi}$ is a vector of length m indicating the number of events per bin. Consequently, when the baseline hazard is approximated by a step function, $\widehat{\sigma}^2_{\widehat{\theta}}$ equals infinity for bins with no event (division by 0). The minimal number of event per bin should then be set to 1.

5.2 Asymptotics for fixed $\frac{\gamma}{1-\gamma}$

The penalised log-likelihood given in (30) may be rewritten under the form of an M-estimator allowing us to obtain, under some regularity conditions, the asymptotic covariance matrix of $\boldsymbol{\eta} = (\boldsymbol{\beta}^T, \boldsymbol{\theta}^T)^T$, $\mathbf{V}(\boldsymbol{\eta})$, by the following sandwich formula [refer to Jun, Héritier and Lô (2013), Section 4.2],

$$\mathbf{V}(\boldsymbol{\eta}) = \mathbf{M}(\boldsymbol{\eta})^{-1}\mathbf{Q}(\boldsymbol{\eta})\mathbf{M}(\boldsymbol{\eta})^{-1}, \tag{63}$$

where

- ▷ $\mathbf{M}(\boldsymbol{\eta}) = \text{E}\left(-\frac{\partial^2}{\partial\boldsymbol{\eta}\partial\boldsymbol{\eta}^T}\boldsymbol{\mathcal{L}}_p\right) = \text{E}\left[-\frac{\partial^2}{\partial\boldsymbol{\eta}\partial\boldsymbol{\eta}^T}(1-\gamma)\boldsymbol{\mathcal{L}} + \frac{\partial^2}{\partial\boldsymbol{\eta}\partial\boldsymbol{\eta}^T}\gamma\mathbf{J}(\boldsymbol{\theta})\right] = \text{E}\left[-(1-\gamma)\boldsymbol{\mathcal{H}} + \frac{\partial^2}{\partial\boldsymbol{\eta}\partial\boldsymbol{\eta}^T}\gamma\mathbf{J}(\boldsymbol{\theta})\right]$, and $\boldsymbol{\mathcal{H}}$ is the hessian matrix of $\boldsymbol{\mathcal{L}}$, the unpenalised likelihood function defined in (32),
- ▷ $\mathbf{Q}(\boldsymbol{\eta}) = \text{E}\left[-\frac{\partial}{\partial\boldsymbol{\eta}}\boldsymbol{\mathcal{L}}_p^T\left(\frac{\partial}{\partial\boldsymbol{\eta}}\boldsymbol{\mathcal{L}}_p^T\right)^T\right]$, where $\boldsymbol{\mathcal{L}}_p = (1-\gamma)\boldsymbol{\mathcal{L}} - \frac{\gamma\mathbf{J}(\boldsymbol{\theta})}{n}\mathbf{1}_n$.

As, for small $\frac{\gamma}{1-\gamma}$ (i.e. $\frac{\gamma}{1-\gamma} < o(\sqrt{n}) = \frac{\sqrt{n}}{10}$, for example), $\widehat{\boldsymbol{\eta}}$ is a consistent estimate of $\boldsymbol{\eta}$, we have that $\sqrt{n}(\widehat{\boldsymbol{\eta}} - \boldsymbol{\eta}) \sim \text{MVN}\left[\mathbf{0}_{p+m}, \mathbf{V}(\boldsymbol{\eta})\right]$, where MVN() denotes the multivariate normal distribution.

The estimated asymptotic covariance matrix of $\widehat{\boldsymbol{\eta}} = (\widehat{\boldsymbol{\beta}}^T, \widehat{\boldsymbol{\theta}}^T)^T$, $\widehat{\mathbf{V}}(\widehat{\boldsymbol{\eta}})$, is then obtained by pluggin-in $\widehat{\boldsymbol{\eta}}$ for $\boldsymbol{\eta}$ in (63) and using the empirical versions of $\mathbf{M}(\boldsymbol{\eta})$ and $\mathbf{Q}(\boldsymbol{\eta})$. We then have

$$\widehat{\mathbf{V}}(\widehat{\boldsymbol{\eta}}) = \widehat{\mathbf{M}}(\widehat{\boldsymbol{\eta}})^{-1}\widehat{\mathbf{Q}}(\widehat{\boldsymbol{\eta}})\widehat{\mathbf{M}}(\widehat{\boldsymbol{\eta}})^{-1}, \tag{64}$$

where $\widehat{\mathbf{M}}(\widehat{\boldsymbol{\eta}})$ and $\widehat{\mathbf{Q}}(\widehat{\boldsymbol{\eta}})$ are (m x m) matrices defined as follows for quadratic penalty functions of form $\mathbf{J}(\boldsymbol{\theta}) = \boldsymbol{\theta}^T\mathbf{R}\boldsymbol{\theta}$ [Note that we propose two ways to define the $\mathbf{M}$ matrix, respectively requiring first and second derivatives of the objective function and denoted by $\mathbf{M}_1$ and $\mathbf{M}_2$]:

$$\widehat{\mathbf{M}}_{\mathbf{1}}(\widehat{\boldsymbol{\eta}}) = -\frac{1}{n}\left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}_p^{\star T} \\ \frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}_p^{\star T}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}} \left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}^{\star T} \\ \frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}^{\star T}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}}^T = -\frac{1}{n}\left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}_p^{\star T} \\ \frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}_p^{\star T}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}} \left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}^T}\boldsymbol{\mathcal{L}}^{\star} \\ \frac{\partial}{\partial\boldsymbol{\theta}^T}\boldsymbol{\mathcal{L}}^{\star}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}}, \tag{65}$$

$$\widehat{\mathbf{M}}_{\mathbf{2}}(\widehat{\boldsymbol{\eta}}) = -\frac{1}{n}\left.\begin{bmatrix} \frac{\partial^2}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^T}\boldsymbol{\mathcal{L}}_p^{\star} & \frac{\partial^2}{\partial\boldsymbol{\beta}\partial\boldsymbol{\theta}^T}\boldsymbol{\mathcal{L}}_p^{\star} \\ \frac{\partial^2}{\partial\boldsymbol{\theta}\partial\boldsymbol{\beta}^T}\boldsymbol{\mathcal{L}}_p^{\star} & \frac{\partial^2}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}\boldsymbol{\mathcal{L}}_p^{\star}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}}, \tag{66}$$

$$\widehat{\mathbf{Q}}(\widehat{\boldsymbol{\eta}}) = -\frac{1}{n}\left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}_p^{\star T} \\ \frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}_p^{\star T}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}} \left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}_p^{\star T} \\ \frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}_p^{\star T}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}}^T = -\frac{1}{n}\left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}_p^{\star T} \\ \frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}_p^{\star T}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}} \left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}^T}\boldsymbol{\mathcal{L}}_p^{\star} \\ \frac{\partial}{\partial\boldsymbol{\theta}^T}\boldsymbol{\mathcal{L}}_p^{\star}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}}. \tag{67}$$

Using the results of appendix Sections A.1 to A.5 and noting that

$$\left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}_p^{\star T} \\ \frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}_p^{\star T}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}} = \left.\begin{bmatrix} (1-\gamma)\frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}^T \\ (1-\gamma)\frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}^T - \frac{\gamma}{n}\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{J}(\boldsymbol{\theta})\mathbf{1}_n^T\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}} \tag{68}$$

$$= \begin{bmatrix} (1-\gamma)_{\mathbb{M}}\left\{_{j=1}^{p} \quad \left[\xi_i - \boldsymbol{\Psi}(t_i)^T\widehat{\boldsymbol{\theta}}e^{\mathbf{x}_i^T\widehat{\boldsymbol{\beta}}}\right]\mathbf{x}_i \quad {}_{i=1}^{n}\right\} \\ (1-\gamma)_{\mathbb{M}}\left\{_{u=1}^{m} \quad \frac{\xi_i}{\boldsymbol{\psi}(t_i)^T\widehat{\boldsymbol{\theta}}}\boldsymbol{\psi}(t_i) - e^{\mathbf{x}_i^T\widehat{\boldsymbol{\beta}}}\boldsymbol{\Psi}(t_i) \quad {}_{i=1}^{n}\right\} - \frac{2\gamma}{n}\mathbf{1}_n \otimes (\mathbf{R}\widehat{\boldsymbol{\theta}})^T\end{bmatrix} \tag{69}$$

$$= \begin{bmatrix} (1-\gamma)\left\{\left[\mathbf{1}_p^T \otimes \left(\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t})\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right)\right] \circ \mathbf{X}\right\}^T \\ (1-\gamma)\left[\left(\mathbf{1}_m^T \otimes \frac{\boldsymbol{\xi}}{\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}}\right) \circ \boldsymbol{\psi}(\mathbf{t}) - \left(\mathbf{1}_p^T \otimes e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right) \circ \boldsymbol{\Psi}(\mathbf{t})\right]^T - \frac{2\gamma}{n}\left(\mathbf{1}_n^T \otimes \mathbf{R}\widehat{\boldsymbol{\theta}}\right)^T\end{bmatrix}, \tag{70}$$

$$\left.\begin{bmatrix} \frac{\partial}{\partial\boldsymbol{\beta}}\boldsymbol{\mathcal{L}}^{\star T} \\ \frac{\partial}{\partial\boldsymbol{\theta}}\boldsymbol{\mathcal{L}}^{\star T}\end{bmatrix}\right|_{\widehat{\boldsymbol{\beta}},\widehat{\boldsymbol{\theta}}} = \begin{bmatrix} (1-\gamma)\left\{\left[\mathbf{1}_p^T \otimes \left(\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t})\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right)\right] \circ \mathbf{X}\right\}^T \\ (1-\gamma)\left[\left(\mathbf{1}_m^T \otimes \frac{\boldsymbol{\xi}}{\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}}\right) \circ \boldsymbol{\psi}(\mathbf{t}) - \left(\mathbf{1}_p^T \otimes e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right) \circ \boldsymbol{\Psi}(\mathbf{t})\right]^T\end{bmatrix}, \tag{71}$$

we obtain the following expressions for $\widehat{\mathbf{M}}_1(\widehat{\boldsymbol{\eta}})$, $\widehat{\mathbf{M}}_2(\widehat{\boldsymbol{\eta}})$

$$\widehat{\mathbf{M}}_{\mathbf{1}}(\widehat{\boldsymbol{\eta}}) = -\frac{1}{n}\begin{bmatrix} (1-\gamma)\left\{\left[\mathbf{1}_p^T \otimes \left(\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t})\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right)\right] \circ \mathbf{X}\right\}^T \\ (1-\gamma)\left[\left(\mathbf{1}_m^T \otimes \frac{\boldsymbol{\xi}}{\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}}\right) \circ \boldsymbol{\psi}(\mathbf{t}) - \left(\mathbf{1}_p^T \otimes e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right) \circ \boldsymbol{\Psi}(\mathbf{t})\right]^T - \frac{2\gamma}{n}\left(\mathbf{1}_n^T \otimes \mathbf{R}\widehat{\boldsymbol{\theta}}\right)^T\end{bmatrix} \tag{72}$$

$$\times \begin{bmatrix} (1-\gamma)\left\{\left[\mathbf{1}_p^T \otimes \left(\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t})\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right)\right] \circ \mathbf{X}\right\}^T \\ (1-\gamma)\left[\left(\mathbf{1}_m^T \otimes \frac{\boldsymbol{\xi}}{\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}}\right) \circ \boldsymbol{\psi}(\mathbf{t}) - \left(\mathbf{1}_p^T \otimes e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right) \circ \boldsymbol{\Psi}(\mathbf{t})\right]^T\end{bmatrix}^T \tag{73}$$

$$\widehat{\mathbf{M}}_{\mathbf{2}}(\widehat{\boldsymbol{\eta}}) = \frac{1}{n}\begin{bmatrix} (1-\gamma)\mathbf{X}^T\left\langle \boldsymbol{\Psi}(t)\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right\rangle\mathbf{X} & (1-\gamma)\mathbf{X}^T\langle e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\rangle\ \boldsymbol{\Psi}(\mathbf{t}) \\ (1-\gamma)\boldsymbol{\Psi}(\mathbf{t})^T\langle e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\rangle\ \mathbf{X} & (1-\gamma)\boldsymbol{\psi}(\mathbf{t})^T\left\langle \boldsymbol{\xi}/[\boldsymbol{\psi}(\mathbf{t})\widehat{\boldsymbol{\theta}}]^2\right\rangle\ \boldsymbol{\psi}(\mathbf{t}) + 2\gamma\mathbf{R}\end{bmatrix}, \tag{74}$$

and $\widehat{\mathbf{Q}}(\widehat{\boldsymbol{\eta}})$

$$\widehat{\mathbf{Q}}(\widehat{\boldsymbol{\eta}}) = -\frac{1}{n}\begin{bmatrix} (1-\gamma)\left\{\left[\mathbf{1}_p^T \otimes \left(\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t})\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right)\right] \circ \mathbf{X}\right\}^T \\ (1-\gamma)\left[\left(\mathbf{1}_m^T \otimes \frac{\boldsymbol{\xi}}{\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}}\right) \circ \boldsymbol{\psi}(\mathbf{t}) - \left(\mathbf{1}_p^T \otimes e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right) \circ \boldsymbol{\Psi}(\mathbf{t})\right]^T - \frac{2\gamma}{n}\left(\mathbf{1}_n^T \otimes \mathbf{R}\widehat{\boldsymbol{\theta}}\right)^T\end{bmatrix} \tag{75}$$

$$\times \begin{bmatrix} (1-\gamma)\left\{\left[\mathbf{1}_p^T \otimes \left(\boldsymbol{\xi} - \boldsymbol{\Psi}(\mathbf{t})\widehat{\boldsymbol{\theta}} \circ e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right)\right] \circ \mathbf{X}\right\}^T \\ (1-\gamma)\left[\left(\mathbf{1}_m^T \otimes \frac{\boldsymbol{\xi}}{\boldsymbol{\psi}(\mathbf{t})\boldsymbol{\theta}}\right) \circ \boldsymbol{\psi}(\mathbf{t}) - \left(\mathbf{1}_p^T \otimes e^{\mathbf{X}\widehat{\boldsymbol{\beta}}}\right) \circ \boldsymbol{\Psi}(\mathbf{t})\right]^T - \frac{2\gamma}{n}\left(\mathbf{1}_n^T \otimes \mathbf{R}\widehat{\boldsymbol{\theta}}\right)^T\end{bmatrix}^T. \tag{76}$$

Note the similarities between $\widehat{\mathbf{M}}_{\mathbf{1}}(\widehat{\boldsymbol{\eta}})$ and $\widehat{\mathbf{Q}}(\widehat{\boldsymbol{\eta}})$, as well as between $\widehat{\mathbf{M}}_{\mathbf{2}}(\widehat{\boldsymbol{\eta}})$ and $\widehat{\mathbf{H}}(\widehat{\boldsymbol{\eta}})$.

The estimated standard deviation vector of $\widehat{\boldsymbol{\eta}}$, $\widehat{\boldsymbol{\sigma}}_{\widehat{\boldsymbol{\eta}}}$, is then given by

$$\widehat{\boldsymbol{\sigma}}_{\widehat{\boldsymbol{\eta}}} = \left\{\frac{1}{n}\mathrm{diag}\left[\widehat{\mathbf{V}}(\widehat{\boldsymbol{\eta}})\right]\right\}^{\frac{1}{2}}. \tag{77}$$

A 1^{st} and 2^{nd} order partial derivatives of the penalised log-likelihood

A.1 $\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}_p^\star = \frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}^\star = \boldsymbol{\nabla}_{\boldsymbol{\beta}}$

$$\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}_p^\star = (1-\gamma)\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L} - \gamma\frac{\partial}{\partial \boldsymbol{\beta}}\boldsymbol{\theta}^T\mathbf{R}\boldsymbol{\theta} \tag{78}$$

$$= (1-\gamma)\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}. \tag{79}$$

As

$$\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}_i = \frac{\partial}{\partial \boldsymbol{\beta}}\left\{\xi_i \log\left[\boldsymbol{\psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\right] - \boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\right\} \tag{80}$$

$$= \xi_i\frac{\boldsymbol{\psi}(t_i)^T\boldsymbol{\theta}}{\boldsymbol{\psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}}\left(\frac{\partial}{\partial \boldsymbol{\beta}}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\right) - \boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta}\left(\frac{\partial}{\partial \boldsymbol{\beta}}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\right) \tag{81}$$

$$= \xi_i\frac{\boldsymbol{\psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}}{\boldsymbol{\psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}}\mathbf{x}_i - \boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\mathbf{x}_i \tag{82}$$

$$= \left[\xi_i - \boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\right]\mathbf{x}_i, \tag{83}$$

where $e^{\mathbf{x}_i^T\boldsymbol{\beta}}$, $\boldsymbol{\psi}(t_i)^T\boldsymbol{\theta} = h_0(t_i)$ and $\boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta} = H_0(t_i)$ are scalars, we have that

$$\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}_p^\star = \nabla_{\boldsymbol{\beta}} = (1-\gamma)\mathbf{X}^T\left[\xi - \boldsymbol{\Psi}(t)\boldsymbol{\theta}\circ e^{\mathbf{X}\boldsymbol{\beta}}\right]. \tag{84}$$

A.2 $\frac{\partial^2}{\partial \boldsymbol{\beta}\partial \boldsymbol{\beta}^T}\mathcal{L}_p^\star = \frac{\partial^2}{\partial \boldsymbol{\beta}\partial \boldsymbol{\beta}^T}\mathcal{L}^\star = \boldsymbol{\mathcal{H}}_{\boldsymbol{\beta}}$

$$\frac{\partial^2}{\partial \boldsymbol{\beta}\partial \boldsymbol{\beta}^T}\mathcal{L}_p^\star = \frac{\partial}{\partial \boldsymbol{\beta}^T}\left(\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}_p^\star\right) = (1-\gamma)\frac{\partial}{\partial \boldsymbol{\beta}^T}\left(\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}\right). \tag{85}$$

As

$$\frac{\partial}{\partial \boldsymbol{\beta}^T}\left(\frac{\partial}{\partial \boldsymbol{\beta}}\mathcal{L}_i\right) = \frac{\partial}{\partial \boldsymbol{\beta}^T}\left(\xi_i - \boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\right)\mathbf{x}_i \tag{86}$$

$$= -\boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta}\left(\frac{\partial}{\partial \boldsymbol{\beta}^T}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\right)\mathbf{x}_i \tag{87}$$

$$= -\boldsymbol{\Psi}(t_i)^T\boldsymbol{\theta}e^{\mathbf{x}_i^T\boldsymbol{\beta}}\mathbf{x}_i^T\mathbf{x}_i, \tag{88}$$

where $e^{\mathbf{x}_i^T\boldsymbol{\beta}}$ and $\boldsymbol{\Psi}^T(t_i)\boldsymbol{\theta}$ are constant scalars, we have that

$$\frac{\partial^2}{\partial \boldsymbol{\beta}\partial \boldsymbol{\beta}^T}\mathcal{L}_p^\star = \boldsymbol{\mathcal{H}}_{\boldsymbol{\beta}} = -(1-\gamma)\mathbf{X}^T\left\langle\boldsymbol{\Psi}(t)\boldsymbol{\theta}\circ e^{\mathbf{X}\boldsymbol{\beta}}\right\rangle\mathbf{X} \tag{89}$$

$$= (\gamma-1)\mathbf{X}^T\left\langle\boldsymbol{\Psi}(t)\boldsymbol{\theta}\circ e^{\mathbf{X}\boldsymbol{\beta}}\right\rangle\mathbf{X}. \tag{90}$$

A.3 $\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L}_p^\star = \boldsymbol{\nabla}_{\boldsymbol{\theta}}$

$$\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L}_p^\star = (1-\gamma) \frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L} - \gamma \frac{\partial}{\partial \boldsymbol{\theta}} \boldsymbol{\theta}^T \mathbf{R} \boldsymbol{\theta} \tag{91}$$

$$= (1-\gamma) \frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L} - 2\gamma \mathbf{R} \boldsymbol{\theta}. \tag{92}$$

As

$$\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L}_i = \frac{\partial}{\partial \boldsymbol{\theta}} \left\{ \xi_i \log \left[\boldsymbol{\psi}(t_i)^T \boldsymbol{\theta} e^{\mathbf{x}_i^T \boldsymbol{\beta}} \right] - \boldsymbol{\Psi}(t_i)^T \boldsymbol{\theta} e^{\mathbf{x}_i^T \boldsymbol{\beta}} \right\} \tag{93}$$

$$= \xi_i \frac{\partial}{\partial \boldsymbol{\theta}} \left\{ \log \left[\boldsymbol{\psi}(t_i)^T \boldsymbol{\theta} e^{\mathbf{x}_i^T \boldsymbol{\beta}} \right] \right\} - e^{\mathbf{x}_i^T \boldsymbol{\beta}} \frac{\partial}{\partial \boldsymbol{\theta}} \left[\boldsymbol{\Psi}(t_i)^T \boldsymbol{\theta} \right] \tag{94}$$

$$= \xi_i \frac{e^{\mathbf{x}_i^T \boldsymbol{\beta}}}{\boldsymbol{\psi}(t_i)^T \boldsymbol{\theta} e^{\mathbf{x}_i^T \boldsymbol{\beta}}} \frac{\partial}{\partial \boldsymbol{\theta}} \left[\boldsymbol{\psi}(t_i)^T \boldsymbol{\theta} \right] - e^{\mathbf{x}_i^T \boldsymbol{\beta}} \frac{\partial}{\partial \boldsymbol{\theta}} \left[\boldsymbol{\Psi}(t_i)^T \boldsymbol{\theta} \right] \tag{95}$$

$$= \frac{\xi_i}{\boldsymbol{\psi}(t_i)^T \boldsymbol{\theta}} \boldsymbol{\psi}(t_i) - e^{\mathbf{x}_i^T \boldsymbol{\beta}} \boldsymbol{\Psi}(t_i), \tag{96}$$

where $e^{\mathbf{x}_i^T \boldsymbol{\beta}}$, $\boldsymbol{\psi}(t_i)^T \boldsymbol{\theta} = h_0(t_i)$ and $\boldsymbol{\Psi}(t_i)^T \boldsymbol{\theta} = H_0(t_i)$ are scalars, we obtain

$$\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L} = \boldsymbol{\psi}(\mathbf{t})^T \left[\boldsymbol{\xi} / \boldsymbol{\psi}(t) \boldsymbol{\theta} \right] - \boldsymbol{\Psi}(\mathbf{t})^T e^{\mathbf{X}\boldsymbol{\beta}}, \tag{97}$$

$$\frac{\partial}{\partial \boldsymbol{\theta}} \boldsymbol{\mathcal{L}}_p^\star = \nabla_{\boldsymbol{\theta}} = (1-\gamma) \left\{ \boldsymbol{\psi}(\mathbf{t})^T \left[\boldsymbol{\xi} / \boldsymbol{\psi}(t) \boldsymbol{\theta} \right] - \boldsymbol{\Psi}(\mathbf{t})^T e^{\mathbf{X}\boldsymbol{\beta}} \right\} - 2\gamma \mathbf{R} \boldsymbol{\theta}. \tag{98}$$

A.4 $\frac{\partial^2}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T} \mathcal{L}_p^\star$

$$\frac{\partial^2}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T} \mathcal{L}_p^\star = \frac{\partial}{\partial \boldsymbol{\theta}^T} \left(\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L}_p^\star \right) = (1-\gamma) \frac{\partial}{\partial \boldsymbol{\theta}^T} \left(\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L} \right) - \frac{\partial}{\partial \boldsymbol{\theta}} 2\gamma \mathbf{R} \boldsymbol{\theta} \tag{99}$$

$$= (1-\gamma) \frac{\partial}{\partial \boldsymbol{\theta}^T} \left\{ \frac{\partial}{\partial \boldsymbol{\theta}} \boldsymbol{\psi}(\mathbf{t})^T \left[\boldsymbol{\xi} / \boldsymbol{\psi}(t) \boldsymbol{\theta} \right] \right\} - 2\gamma \mathbf{R} \tag{100}$$

$$= (1-\gamma) \frac{\partial}{\partial \boldsymbol{\theta}^T} \left\{ \boldsymbol{\psi}(\mathbf{t})^T \left[\boldsymbol{\xi} / \boldsymbol{\psi}(t) \boldsymbol{\theta} \right] \right\} - 2\gamma \mathbf{R} \tag{101}$$

$$= (1-\gamma) \boldsymbol{\psi}(\mathbf{t})^T \frac{\partial}{\partial \boldsymbol{\theta}^T} \left[\boldsymbol{\xi} \circ \frac{1}{\boldsymbol{\psi}(t) \boldsymbol{\theta}} \right] - 2\gamma \mathbf{R} \tag{102}$$

$$= (1-\gamma) \boldsymbol{\psi}(\mathbf{t})^T \; \boldsymbol{\xi} \circ \frac{\partial}{\partial \boldsymbol{\theta}^T} \left[\frac{1}{\boldsymbol{\psi}(t) \boldsymbol{\theta}} \right] - 2\gamma \mathbf{R} \tag{103}$$

$$= (\gamma-1) \boldsymbol{\psi}(\mathbf{t})^T \left\langle \boldsymbol{\xi} / [\boldsymbol{\psi}(\mathbf{t}) \boldsymbol{\theta}]^2 \right\rangle \boldsymbol{\psi}(\mathbf{t}) - 2\gamma \mathbf{R}. \tag{104}$$

A.5 $\frac{\partial^2}{\partial \boldsymbol{\theta} \partial \boldsymbol{\beta}^T} \mathcal{L}_p^\star = \frac{\partial^2}{\partial \boldsymbol{\theta} \partial \boldsymbol{\beta}^T} \mathcal{L}^\star$ and $\frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\theta}^T} \mathcal{L}_p^\star = \frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\theta}^T} \mathcal{L}^\star$

$$\frac{\partial^2}{\partial \boldsymbol{\theta} \partial \boldsymbol{\beta}^T} \mathcal{L}_p^\star = \frac{\partial}{\partial \boldsymbol{\beta}^T} \left(\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L}_p^\star \right) = (1-\gamma) \frac{\partial}{\partial \boldsymbol{\beta}^T} \left(\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{L} \right) \tag{105}$$

$$= (1-\gamma) \frac{\partial}{\partial \boldsymbol{\beta}^T} \left[-\boldsymbol{\Psi}(\mathbf{t})^T e^{\mathbf{X}\boldsymbol{\beta}} \right] \tag{106}$$

$$= (\gamma-1) \boldsymbol{\Psi}(\mathbf{t})^T \frac{\partial}{\partial \boldsymbol{\beta}^T} e^{\mathbf{X}\boldsymbol{\beta}} \tag{107}$$

$$= (\gamma-1) \boldsymbol{\Psi}(\mathbf{t})^T \left\langle e^{\mathbf{X}\boldsymbol{\beta}} \right\rangle \mathbf{X}, \tag{108}$$

$$\frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\theta}^T} \mathcal{L}_p^\star = \left(\frac{\partial^2}{\partial \boldsymbol{\theta} \partial \boldsymbol{\beta}^T} \mathcal{L}_p^\star \right) = (\gamma-1) \mathbf{X}^T \left\langle e^{\mathbf{X}\boldsymbol{\beta}} \right\rangle \boldsymbol{\Psi}(\mathbf{t}). \tag{109}$$